



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS4130 – Computer Network Laboratory

Experiment No. 02

Introduction to Cisco Packet Tracer – Static Routing

■ Objectives

- (a) Gain familiarity with the Cisco Packet Tracer network simulator.
- (b) Learn how to configure a Cisco router using the command-line interface (CLI).
- (c) Understand and configure static routing to enable communication between different networks.
- (d) Learn how to use a packet sniffer to capture and analyze network traffic.

■ Lab Requirements

- (a) Two Cisco routers.
- (b) Two Cisco switches.
- (c) One sniffer.
- (d) Two PCs.

■ Pre-Lab

- (a) Install Cisco Packet Tracer (latest version) on your personal computer.
- (b) Review IP subnetting concepts and work through several practice examples.

2.1 Introduction

In the previous experiment, you built simple single-segment networks and used the TCP/IP protocol suite to enable communication among the devices within each network. However, when multiple networks need to be interconnected, a device capable of forwarding data packets from one network to the correct destination network is required. This device is called a **router**, and its primary function is to route packets toward their intended destination.

Traditionally, a router is referred to as a **Layer-3 device** because it operates at the Network Layer of the OSI model and relies on IP addresses (Layer-3 addresses) to determine the forwarding path toward a destination. Each individual network connected to a router is called a **segment** (or **subnet**). The primary purpose of subnetting is to control and manage network traffic; every node within a given segment can observe all packets transmitted by other nodes on that same segment.

Using routing information stored in a **routing table**, a router determines the next hop toward the destination for each incoming packet. The router examines the destination IP address contained in the packet header to identify the correct forwarding path.

There are two main categories of routing: **static** and **dynamic**. In static routing, the network administrator is responsible for manually entering and updating all routing information (e.g., adding or removing routes to network segments). In dynamic routing, routing protocols automatically exchange network reachability information among routers and update their routing tables accordingly.

2.2 Routing

Routing refers to the process of forwarding a packet from a device on one network to a device on a different network. Routers are not concerned with individual hosts; their sole purpose is to deliver packets to the correct destination *network*. Once a packet arrives at the destination network, the hardware (MAC) address of the target host is used to deliver the packet to its final destination. If your network contains no routers, then no routing is taking place.

To route packets successfully, a router must know, at a minimum, the following:

- The destination network address.
- The neighboring routers from which it can learn about remote networks.
- All possible routes to each remote network.
- The best (preferred) route to each remote network.
- How to maintain and verify its routing information.

2.2.1 Static Routing & Dynamic Routing

A router learns about remote networks either from neighboring routers or from a network administrator. Based on this information, the router constructs a routing table that describes how to reach each known network. If a network is directly connected to one of the router's interfaces, the router inherently knows how to reach it. For networks that

are not directly connected, the router must learn the path in one of two ways: through **static routing**, in which an administrator manually enters all network routes into the routing table, or through **dynamic routing**, in which routing protocols automatically discover and propagate route information.

A) Static Routing

Static routing involves manually adding routes to each router's routing table.

- **Benefits of static routing:**

- There is no additional overhead on the router's CPU, which means a less powerful (and less expensive) router can be used compared to one running dynamic routing protocols.
- There is no bandwidth consumption between routers for exchanging routing updates, which can result in cost savings on WAN links.
- It provides an added layer of security, as the administrator can selectively control which networks are reachable by configuring only the desired routes.

- **Disadvantages of static routing:**

- The administrator must have a thorough understanding of the entire inter-network topology and how each router is interconnected in order to configure routes correctly.
- Whenever a new network is added to the internetwork, the administrator must manually add a corresponding route on every affected router.
- It is not scalable for large networks, as maintaining static routes across a large number of routers becomes impractical and error-prone.

B) Dynamic Routing

In dynamic routing, a routing protocol running on one router communicates with the same protocol running on neighboring routers. The routers exchange information about all the networks they are aware of and update their routing tables accordingly. If a topology change occurs (e.g., a link goes down or a new network is added), the dynamic routing protocol automatically propagates the change to all participating routers. In contrast, with static routing, the administrator must manually update every affected router. In practice, large networks typically employ a combination of both static and dynamic routing.

2.3 Router Configuration

Router configuration involves setting up interfaces, assigning IP addresses, and configuring routing to enable communication between networks. The following subsections describe this process step by step, starting from router initialization through to static routing setup.

2.3.1 Setup Mode and CLI Mode

When a router initializes, it sequentially loads the bootstrap program, the operating system (IOS), and the startup configuration file. If no configuration file is found, the router enters **setup mode**, which provides a guided configuration wizard.

- To exit setup mode at any time, press **Ctrl+Z**.
- The CLI (Command-Line Interface) is the most flexible and powerful method for configuring a Cisco router.
- To bypass setup mode entirely, type **no** when prompted with the configuration dialog.

```
Continue with configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>
```

2.3.2 Logging into the Router

After the interface status messages are displayed and you press **Enter**, the **Router>** prompt appears. This is called **user EXEC mode** (also known as user mode) and is primarily used for viewing statistics and basic monitoring. To access full configuration capabilities, you must enter **privileged EXEC mode** using the **enable** command:

```
Router> enable
Router#
```

The **Router#** prompt indicates that you are in privileged mode, where configuration changes can be made. To return to user EXEC mode, use the **disable** command:

```
Router# disable
Router>
```

To log out of the console entirely, type **logout**:

```
Router> logout
```

2.3.3 Overview of Router Modes

To begin configuring the router, enter **global configuration mode** by typing **configure terminal** (abbreviated as **config t**):

```
Router# configure terminal
Router(config)#
```

In global configuration mode, commands affect the router's overall operation. To configure specific interfaces, routing protocols, or other features, you must enter the appropriate sub-configuration mode from within global configuration mode.

2.3.4 Router Interfaces

To configure a specific router interface, use the **interface** command from global configuration mode:

```
Router(config)# interface <TYPE> <SLOT>/<PORT>
```

For example, to enter the configuration mode for interface **FastEthernet 0/0**:

```
Router(config)# interface FastEthernet 0/0
Router(config-if)#
```

Alternatively, you may use the abbreviated form:

```
Router(config)# interface fa0/0
Router(config-if)#
```

Interface configuration is essential because, without properly configured and active interfaces, a router cannot communicate with any network. To exit interface configuration mode and return to global configuration mode:

```
Router(config-if)# exit
Router(config)#
```

2.3.5 Configuring Router Interfaces

Each router interface must be assigned an IP address and a subnet mask to enable proper communication. The syntax for assigning an IP address to an interface is:

```
Router(config-if)# ip address <IP-ADDRESS> <SUBNET-MASK>
```

2.3.6 Bringing Up an Interface

By default, router interfaces may be in an administratively down state. To enable (bring up) an interface, use the **no shutdown** command:

```
Router(config-if)# no shutdown
```

To administratively disable an interface at a later time, use the **shutdown** command:

```
Router(config-if)# shutdown
```

2.3.7 Configuring Static Routing

Once all interfaces are active and IP addresses have been assigned, the router needs routes to reach remote (non-directly-connected) networks. Static routing allows the administrator to manually define these network paths.

The syntax for adding a static route to the routing table is:

```
Router(config)# ip route <DESTINATION-NETWORK-ID>
                    <DESTINATION-SUBNET-MASK>
                    <NEXT-HOP-ADDRESS>
```

The parameters are defined as follows:

- **ip route:** The command used to create a static route entry.
- **DESTINATION-NETWORK-ID:** The network address of the remote network to be added to the routing table.
- **DESTINATION-SUBNET-MASK:** The subnet mask associated with the destination network.
- **NEXT-HOP-ADDRESS:** The IP address of the next-hop router that will receive and forward the packet toward the destination network. This address must belong to a directly connected network.

Important: You must be able to successfully ping the next-hop router interface before adding the static route. If an incorrect next-hop address is specified, or if the interface to the next-hop router is down, the static route will appear in the router's configuration file but will *not* be installed in the routing table.

2.4 Packet Sniffing

Packet sniffing is a technique used to monitor and analyze network traffic by capturing data packets as they traverse the network. Each captured packet contains valuable information, including source and destination IP addresses, the protocols in use, and the actual data payload.

Packet sniffers—such as Wireshark or the Simulation Mode in Cisco Packet Tracer—capture network packets and allow users to inspect their contents in detail. The general process works as follows:

- **Capturing Packets:** The sniffer listens to network traffic by placing a network interface in *promiscuous mode* (on wired networks) or by monitoring wireless signals (on Wi-Fi networks), enabling it to capture all passing traffic rather than only traffic addressed to the local host.
- **Decoding Data:** Once captured, the tool deconstructs each packet into its constituent parts, displaying readable details such as headers, payloads, and protocol information at each layer.
- **Analyzing Network Activity:** Users can apply filters to isolate specific types of traffic, detect errors, and analyze communication patterns between devices.

2.4.1 Why Is Packet Sniffing Useful?

Packet sniffing is widely used in IT and cybersecurity for several purposes:

- **Troubleshooting Network Issues** – Helps identify slow connections, dropped packets, and misconfigured devices.
- **Security Monitoring** – Detects suspicious activity, unauthorized data transfers, and potential cyber threats.
- **Protocol Analysis** – Examines how different network protocols operate and verifies that they function correctly.

Cisco Packet Tracer includes a **Simulation Mode** that allows users to inspect packets step by step as they move through the network. This feature enables students and professionals to:

- **Visualize Data Flow:** Observe how packets travel between devices across the network.
- **Inspect Packet Details:** Click on a captured packet to view detailed information, including the source address, destination address, and protocol used.
- **Understand Protocol Operations:** Analyze the different protocol layers to learn how data is encapsulated, transmitted, and processed at each stage.

2.5 Procedure

In this lab, we will interconnect two routers and two PCs located on different networks. This requires configuring a routing mechanism between the routers to enable cross-network communication. Specifically, we will configure **static routing** on both routers.

2.5.1 Building the Topology

Construct the network topology shown in Figure 1, which consists of the following devices:

- **Routers:** Use Router-PT.
- **Switches:** Use Switch-PT.
- **PCs:** Use PC-PT.
- **Packet Capture:** Place a Sniffer between **Router0** and **Switch0** to monitor incoming packets effectively.
- For all connections between PCs, switches, and routers, use **Automatically Choose Connection Type**.

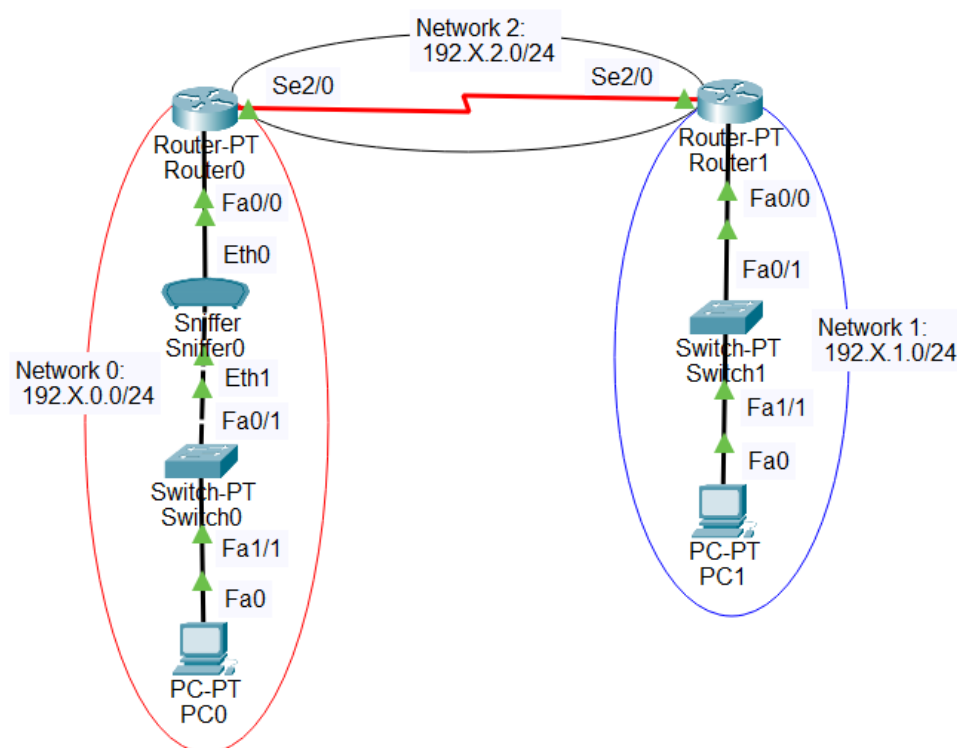


Figure 1: Network topology.

Assign appropriate IP addresses to all PCs and routers, ensuring that each device receives a unique address within the valid host range of its respective subnet. The value of **X** in the network address corresponds to the *last two digits* of your student ID. For example, if the network address is **192.X.1.0/24** and your student ID is **1230105**, then **X = 05**, resulting in the network address **192.5.1.0/24**.

2.5.2 Configuring PCs

- Click on **PC0** and navigate to the Desktop tab.
- Select IP Configuration and assign the IP address 192.X.0.2/24.
- Repeat the same procedure for **PC1**, assigning it the IP address 192.X.1.2/24.
- **Note:** A subnet mask of /24 is equivalent to 255.255.255.0. Figure 2 shows the IP configuration for **PC0**.

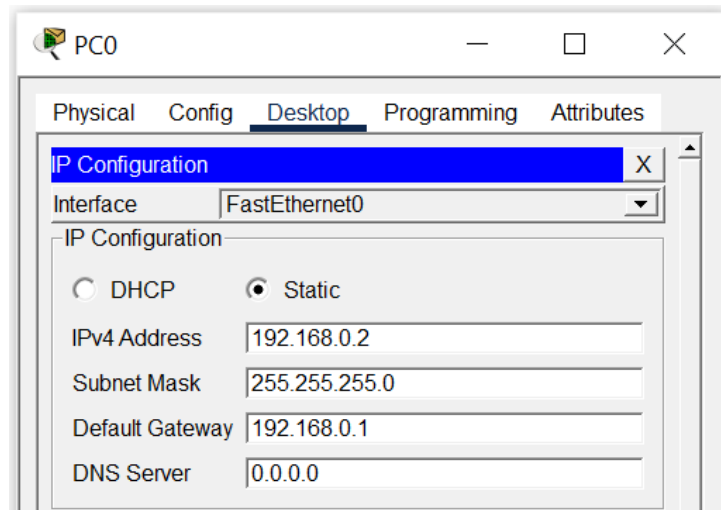


Figure 2: IP configuration of PC0.

2.5.3 Configuring Routers

A) Assigning IP Addresses to Router Interfaces

- Click on **Router0** and navigate to the CLI tab, as shown in Figure 3.

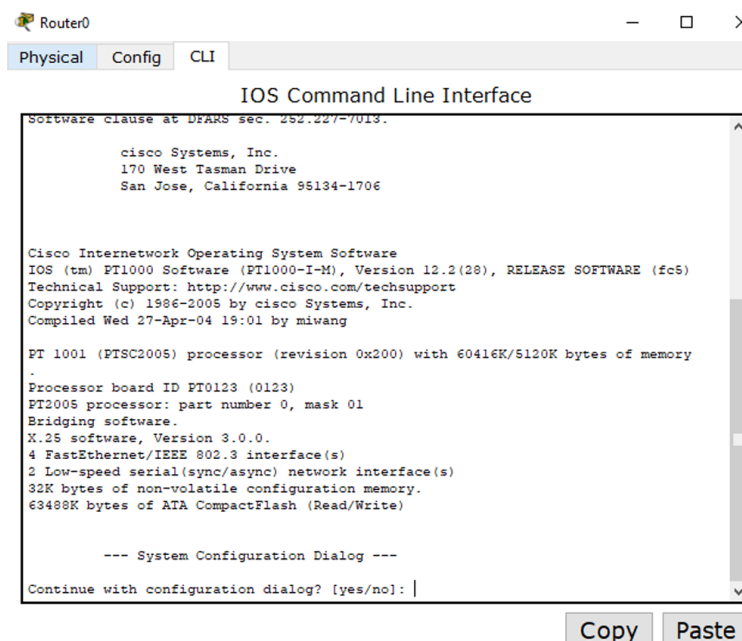


Figure 3: CLI of Router0.

Before beginning the configuration, enter the following commands to access global configuration mode:

```
Router> enable
Router# configure terminal
Router(config)#
```

The IP addresses and subnet masks for each router interface are listed in Table 1.

Table 1: Router interface IP configuration.

Router	Interface	IP Address	Subnet Mask
Router0	Fa0/0	192.X.0.1	255.255.255.0
	Se2/0	192.X.2.1	255.255.255.0
Router1	Fa0/0	192.X.1.1	255.255.255.0
	Se2/0	192.X.2.2	255.255.255.0

To configure the IP address on **Router0** for interface **Fa0/0**:

```
Router(config)# interface Fa0/0
Router(config-if)# ip address 192.X.0.1 255.255.255.0
Router(config-if)# no shutdown
```

Repeat the above steps for **Router0** interface **Se2/0** and **Router1** interfaces **Fa0/0** and **Se2/0**, using the corresponding IP addresses from Table 1.

B) Configuring Static Routing

To configure a static route on **Router0** so that it can reach network 192.X.1.0/24 (the network behind **Router1**):

```
Router(config)# ip route 192.X.1.0 255.255.255.0 192.X.2.2
```

Repeat the above command on **Router1**, specifying the appropriate destination network and next-hop address to reach network 192.X.0.0/24.

2.5.4 Troubleshooting and Verifying the Configuration

After completing the static routing configuration, it is essential to verify that all routers and end devices can communicate properly. The following steps and commands assist in inspecting configurations, verifying routing behavior, and troubleshooting potential issues.

A) Verifying End-to-End Connectivity

Once all routers have been configured with static routes, you must verify connectivity across the entire network. The most effective method—in addition to inspecting the routing table with the `show ip route` command—is to use the `ping` command to test end-to-end connectivity.

To verify connectivity between end devices:

1. Open **PC0**.
2. Navigate to **Desktop** → **Command Prompt**.
3. Enter the following command:

```
ping <IP-ADDRESS>  
Ex: ping 192.X.1.2
```

If the ping replies are successful, as shown in Figure 4, then the static routes and interface configurations are correct.

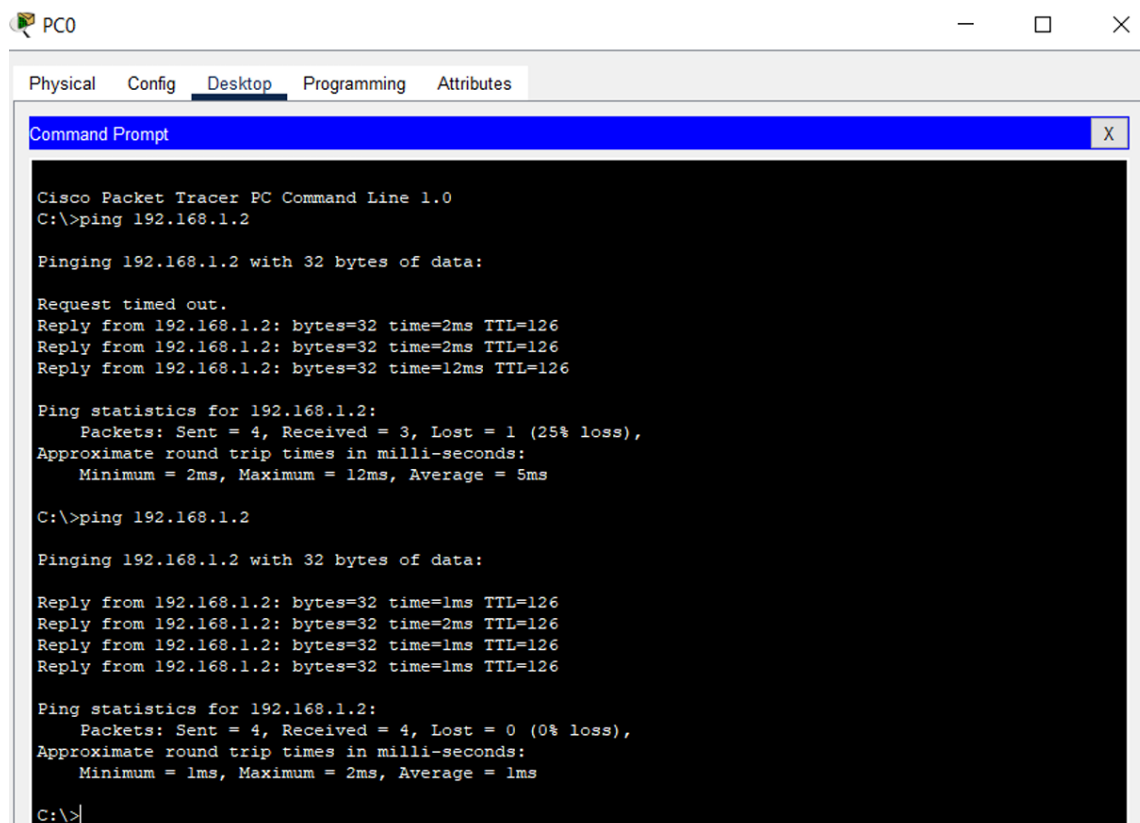


Figure 4: Pinging to test end-to-end connectivity.

B) Displaying and Verifying the Running Configuration

To confirm that all static routes and interface settings have been correctly applied, use the following command:

```
Router# show running-config
```

This command displays the router's current (active) configuration, including all interface settings and manually configured static routes, as shown in Figure 5.

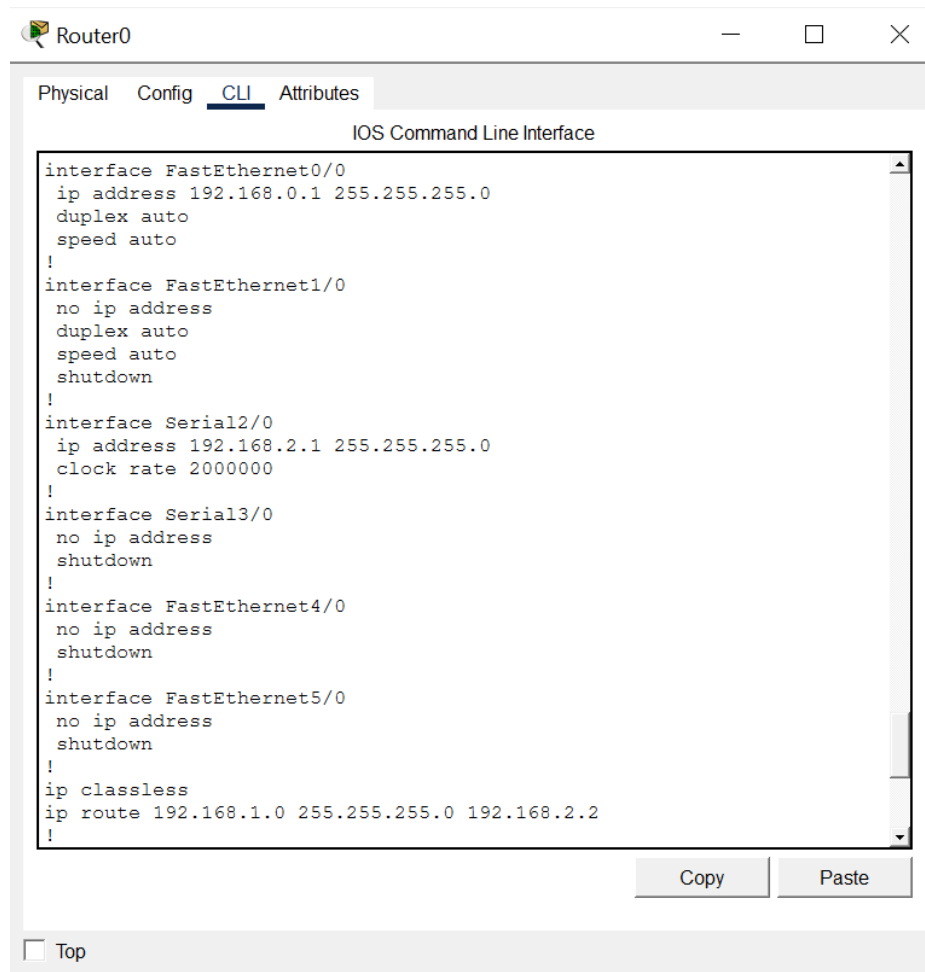


Figure 5: Running configuration on Router0.

C) Displaying the Routing Table

Static routes are manually added to the routing table by the network administrator. To verify that these routes have been correctly installed, use:

```
Router# show ip route
```

This command lists all known routes, including:

- **C** — Directly connected networks.
- **S** — Static routes manually configured by the administrator.

In the routing table output shown in Figure 6 for **Router0**, each static route entry is prefixed with the letter **S** and displays the destination network, the next-hop IP address, and the outgoing interface. For example, the entry:

```
S      192.X.1.0/24 [1/0] via 192.X.2.2
```

indicates that the router can reach network 192.X.1.0/24 via the next-hop address 192.X.2.2 through the Serial2/0 interface. The notation [1/0] represents an administrative distance of 1 (the default for static routes) and a metric of 0.

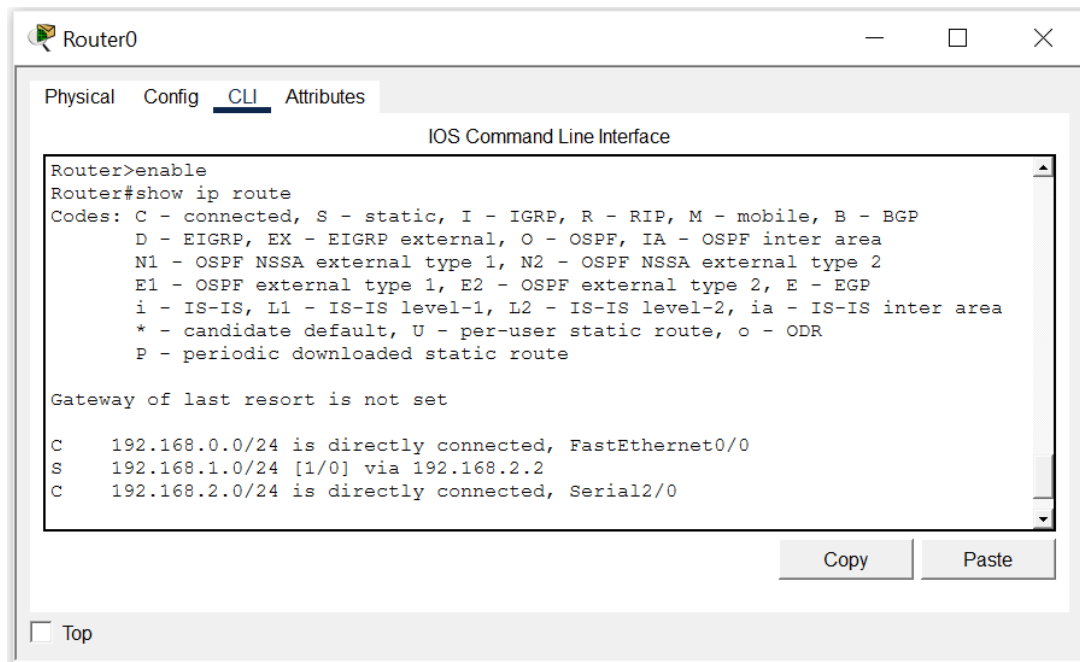


Figure 6: Routing table on Router0 (static routing).

D) Checking Interface Status and IP Address Configuration

Before troubleshooting routing issues, ensure that all router interfaces are properly configured and operational. Use the following command:

```
Router# show ip interface brief
```

This command provides a concise summary of all interfaces, displaying their assigned IP addresses, operational status (up/down), and line protocol status.

As shown in Figure 7, **Router1** has interfaces *FastEthernet0/0* and *Serial2/0* both in the “up/up” state, indicating that the interfaces are active and functioning properly.

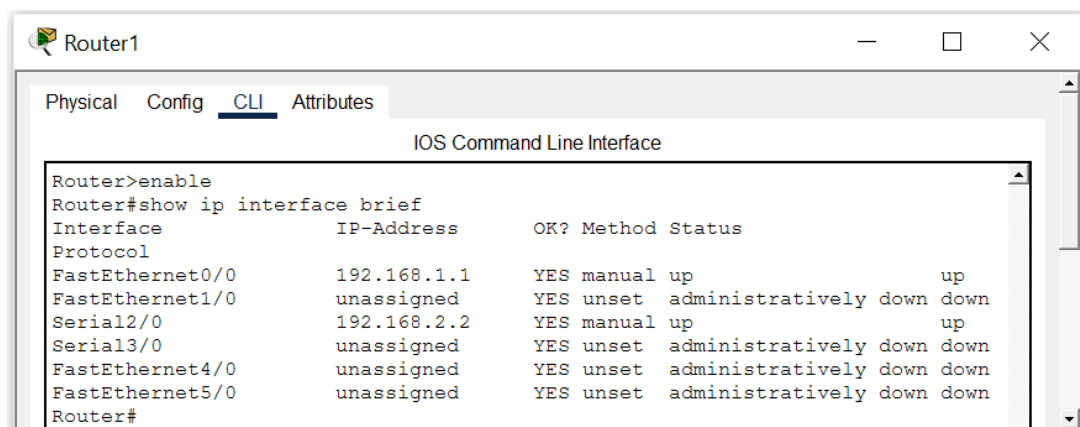


Figure 7: Interface summary on Router1.

2.5.5 Securing Router Access with Passwords

Passwords are essential for controlling access to a Cisco router. They determine who can log in and what level of access each user has. There are three primary types of passwords that should be configured:

1. Console Password

The console password restricts access when a user connects physically to the router through the console port. Without this password, anyone with physical access to the device could modify its configuration.

```
Router(config)# line console 0
Router(config-line)# password 111
Router(config-line)# login
```

The `line console 0` command enters console line configuration mode. The `password 111` command sets the console password (in this example, 111). Finally, the `login` command enables password checking, causing the router to prompt for the password upon console access.

2. Privileged (Enable) Password

The privileged password protects access to privileged EXEC mode, where advanced configuration commands are available. There are two methods for setting this password:

```
Router(config)# enable password 222
```

The `enable password` command sets a plain-text password. However, this method is *not secure*, as the password is stored in clear text and can be easily read from the configuration file.

```
Router(config)# enable secret 222
```

The `enable secret` command sets an encrypted password using a one-way hash algorithm (MD5) and is the recommended option for securing privileged access.

Note:

If both `enable secret` and `enable password` are configured, the `enable secret` always takes precedence and overrides the `enable password`.

3. Telnet (VTY) Password

The Telnet password secures remote access to the router through virtual terminal (VTY) lines. This prevents unauthorized users from connecting to the router over the network.

```
Router(config)# line vty 0 4
Router(config-line)# password 333
Router(config-line)# login
```

The `line vty 0 4` command enters configuration mode for VTY lines 0 through 4, allowing up to five simultaneous remote (Telnet) sessions. The `password 333`

command sets the VTY password (in this example, 333). The `login` command activates password protection for remote access.

2.5.6 Packet Sniffing

Packet sniffing allows us to monitor and analyze network traffic by capturing packets as they travel between devices. In Cisco Packet Tracer, the Sniffer tool is used to visualize packet movement and examine how data flows through the network.

To configure the sniffer, navigate to **Sniffer** → **GUI** → **Service** and toggle packet capturing **On** or **Off**. Then, select **Port0** or **Port1** to specify which interface will be used to capture incoming packets. This setting allows you to control the direction and source of the captured traffic.

By default, the sniffer captures all protocols. However, you can refine the capture by:

- Clicking **Clear** to remove all previously captured packets.
- Clicking **Edit Filters** to select specific protocols for more targeted analysis.

Figure 8 shows the sniffer capturing all packets traversing the network.

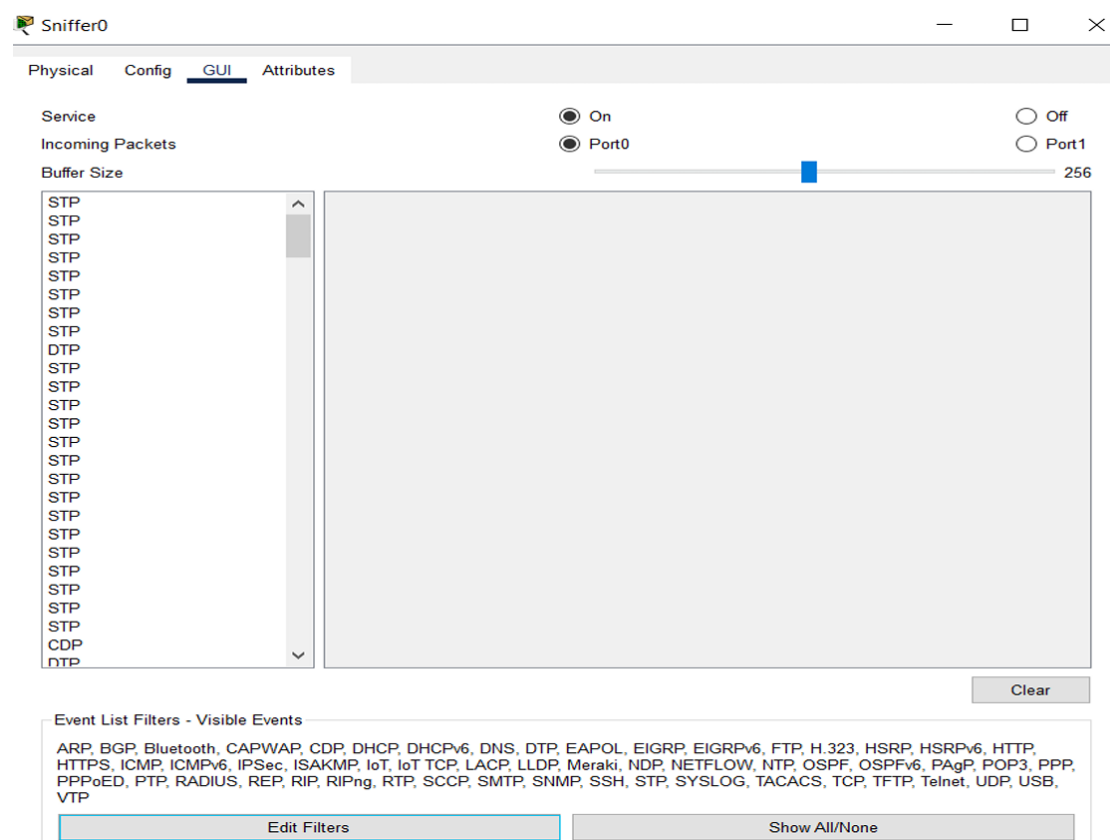


Figure 8: Capturing all packets.

Since we are testing connectivity using the `ping` command (which relies on the ICMP protocol), we apply a filter to display only ICMP packets. This makes it easier to verify whether the ping was successful without being distracted by other traffic. Figure 9 illustrates how to apply a protocol-specific filter.

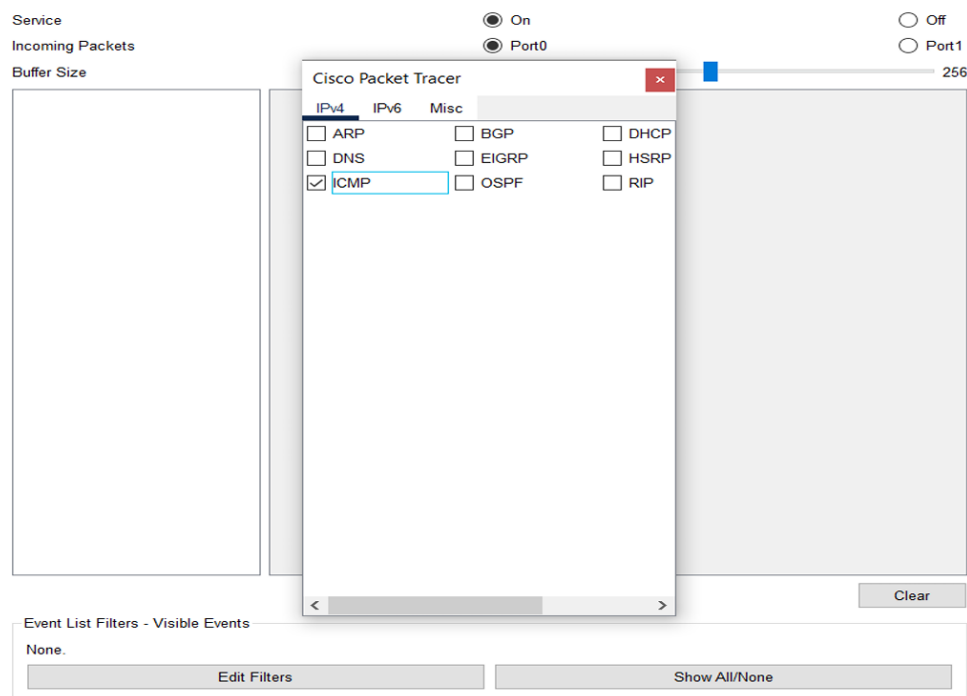


Figure 9: Applying a filter to capture a specific protocol.

After applying the ICMP filter and sending a ping from **PC0** to **PC1**, we can click on a captured packet in the sniffer tool to examine its details. The packet details include the following information, as shown in Figure 10:

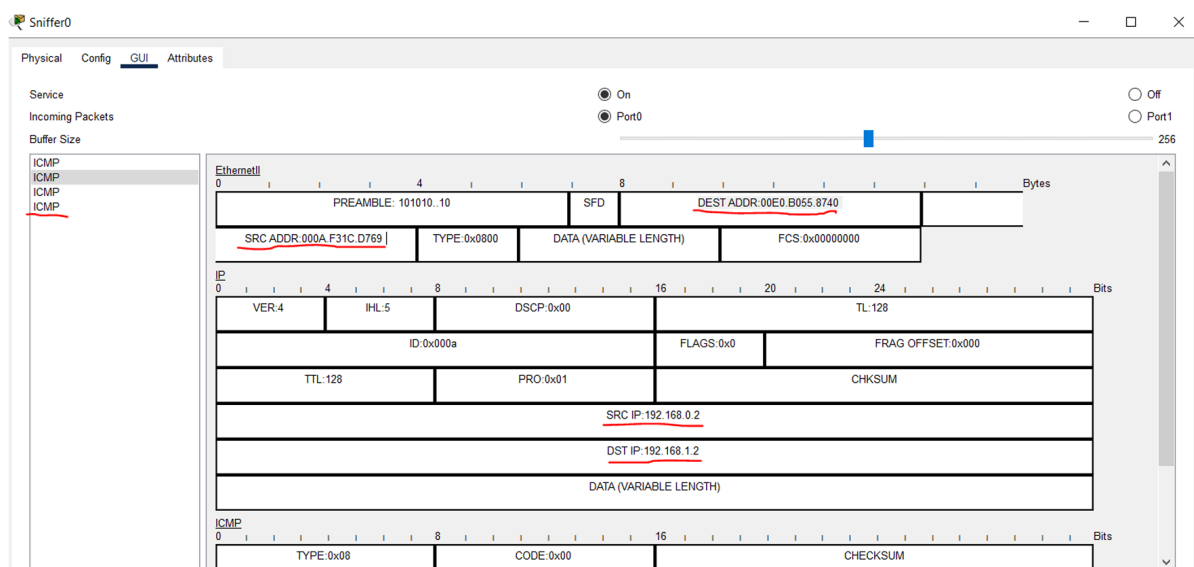


Figure 10: Packet details.

- **Source MAC Address:** The unique hardware address of the device that initiated the packet transmission—in this case, **PC0**. The MAC address operates at the Data Link Layer (Layer 2) and is used to identify the sending device on the local network segment.
- **Destination MAC Address:** The hardware address of the next device the packet

is forwarded to—in this case, **Router0's Fa0/0 interface** (not **PC1**, since the packet must first reach the router to be forwarded to the remote network). The destination MAC address ensures the frame reaches the correct next-hop device on the local network.

- **Source IP Address:** The IP address of the sender, which is **PC0**. The IP address operates at the Network Layer (Layer 3) and is used for end-to-end routing of the packet across multiple networks.
- **Destination IP Address:** The IP address of the intended recipient, which is **PC1**. Routers and other Layer-3 devices use this address to determine the final destination of the packet across different network segments.

To verify the source MAC address, click on **PC0**, navigate to the **Config** tab, select **FastEthernet**, and view the **MAC Address** field. The MAC address displayed should match the source MAC address in the captured packet, confirming that the packet originated from **PC0**. Figure 11 shows the source MAC address for **PC0**.

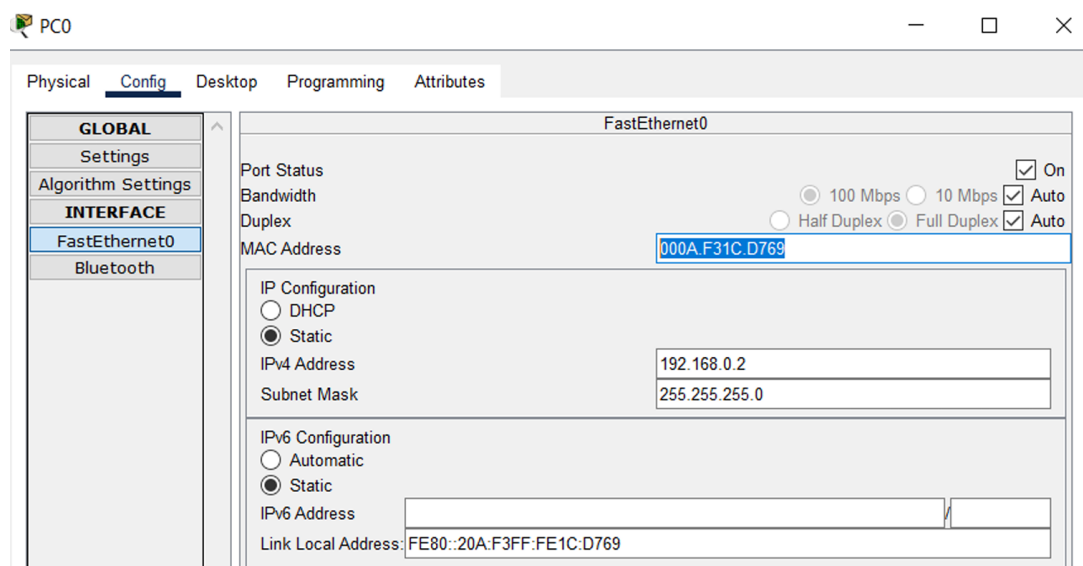


Figure 11: Source MAC address for PC0.

2.6 Other Important Configurations

2.6.1 Editing and Help Features

The Cisco IOS provides built-in help features that assist with command entry. Typing a question mark (?) at any prompt displays the available commands or parameters. To list all commands that begin with a specific letter, type the letter followed by ? with no space between them:

```
Router# c?
Clear clock configures connect copy
```

By typing `c?`, the router displays all commands starting with `c`. To explore the next parameter in a command sequence, type the command followed by a space and then `?`:

```

Router#clock ?
Set the time and date

Router#clock set ?
hh:mm:ss Current Time

Router#clock set 10:30:10 ?
<1-31> Day of the month
MONTH Month of the year

Router#clock set 10:30:10 28 ?
MONTH Month of the year

Router#clock set 10:30:10 28 january ?
<1993-2035> Year

Router#clock set 10:30:10 28 january 2020 ?
<cr>

```

By using the `clock ?` command, the router presents the available parameters at each stage along with brief descriptions. Continue entering each parameter followed by a space and `?` until the only remaining option is `<cr>` (carriage return), indicating that the command is complete.

In addition to the context-sensitive help, the Cisco IOS provides a set of keyboard short-cuts that streamline command-line editing and navigation. Table 2 summarizes the most commonly used enhanced editing commands.

Table 2: Enhanced editing commands.

Command	Description
Ctrl+P or Up Arrow	Recalls the previous (most recent) command
Ctrl+N or Down Arrow	Recalls the next (older) command in the history
<code>show history</code>	Displays the last 10 commands entered (by default)
Ctrl+A	Moves the cursor to the beginning of the line
Ctrl+E	Moves the cursor to the end of the line
Esc+B	Moves the cursor back one word
Ctrl+F	Moves the cursor forward one character
Esc+F	Moves the cursor forward one word
Ctrl+B	Moves the cursor back one character
Ctrl+D	Deletes the character at the cursor
Backspace	Deletes the character before the cursor
Ctrl+R	Redisplays the current command line
Ctrl+U	Erases the entire line
Ctrl+W	Erases the word to the left of the cursor
Ctrl+Z	Exits configuration mode and returns to privileged EXEC
Tab	Auto-completes a partially typed command

2.6.2 Gathering Basic Router Information

The `show version` command displays fundamental information about the router, including the hardware platform, the IOS software version, the configuration register setting, and the sources of the boot image and configuration files:

```
Router# show version
```

2.6.3 Serial Interface Commands

When configuring serial interfaces, there are several important considerations. In production environments, the serial interface is typically connected to a CSU/DSU device, which provides the clocking signal for the link. However, in a lab environment with a back-to-back serial connection, one end of the cable—the **DCE (Data Communication Equipment)** end—must provide the clocking signal.

By default, all Cisco router serial interfaces operate as **DTE (Data Terminal Equipment)** devices. Therefore, if a router's serial interface is connected to the DCE end of the cable, you must configure it to provide clocking using the `clock rate` command:

```
Router# config t
Router(config)# int se2/0
Router(config-if)# clock rate ?
Speed (bits per second
  1200
  2400
  4800
  9600
  19200
  38400
  56000
  64000
  72000
  125000
  128000
  148000
  250000
  500000
  800000
  1000000
  1300000
  2000000
  4000000
  <300-4000000> Choose clockrate from list above
Router(config-if)# clock rate 64000
```

The `bandwidth` command is used to set the logical bandwidth value on the interface (used by routing protocols for metric calculation, not to limit actual throughput):

```
Router(config-if)# bandwidth <BANDWIDTH-IN-KILOBITS>
Router(config-if)# bandwidth 64
```

2.6.4 Hostnames

You can assign a descriptive name to the router using the **hostname** command. This name is only locally significant—it does not affect how the router performs name resolution or how it operates within the internetwork. It does, however, appear in the CLI prompt and aids in identifying the device:

```
Router# config t
Router(config)# hostname RouterA
RouterA(config)# hostname RouterB
RouterB(config)#
```

It is best practice to assign a hostname that reflects the router's physical location or function within the network (e.g., **HQ-Router**, **Branch-01**), rather than a personal name.

2.6.5 Viewing and Saving Configurations

When exiting setup mode, the router will ask whether you want to save the configuration you just created. If you answer **Yes**, the router copies the configuration currently running in DRAM (known as the **running-config**) into NVRAM, where it is saved as the **startup-config** file.

You can manually save the running configuration to NVRAM at any time using:

```
Router# copy running-config startup-config
```

(The shorthand **copy run start** is also accepted.)

Alternatively, you can use the legacy command:

```
Router# write
```

Saving the configuration ensures that your changes persist across router reboots and power cycles.

2.7 ToDo

This section will be provided by the instructor.